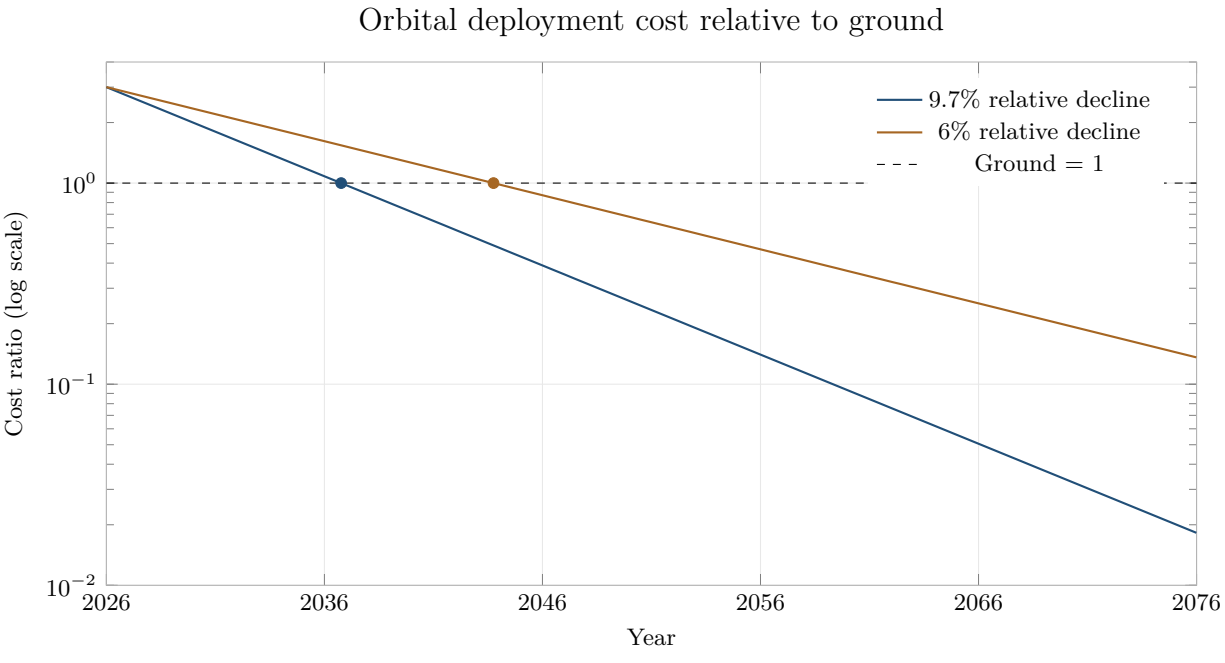
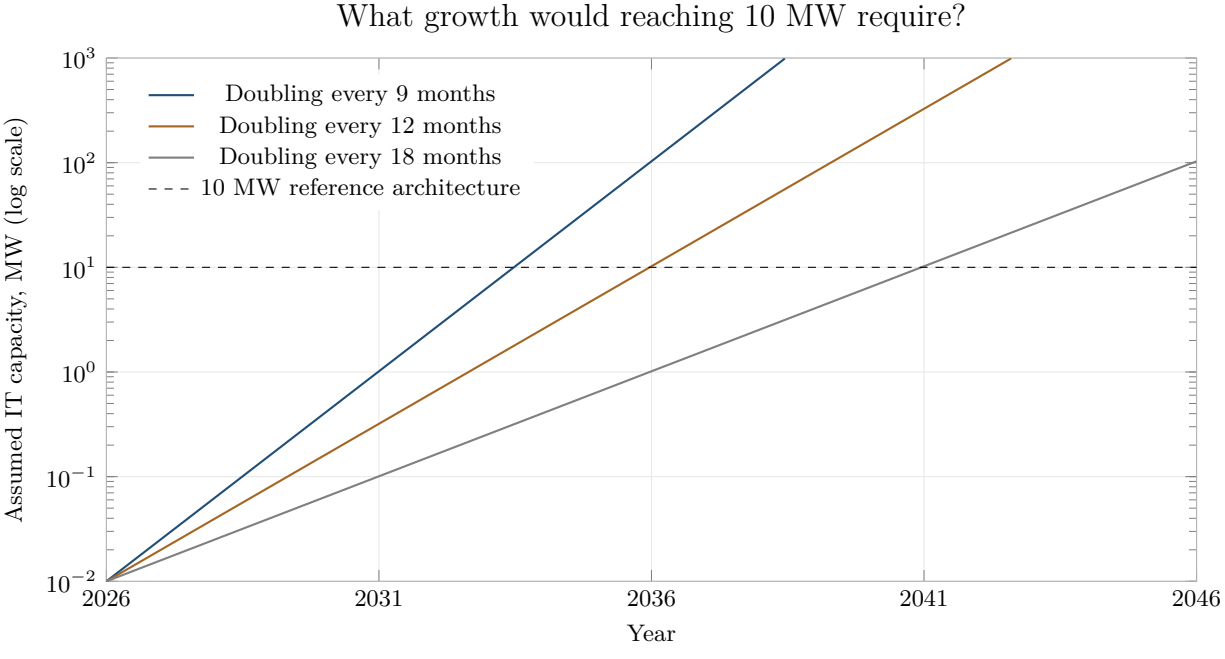
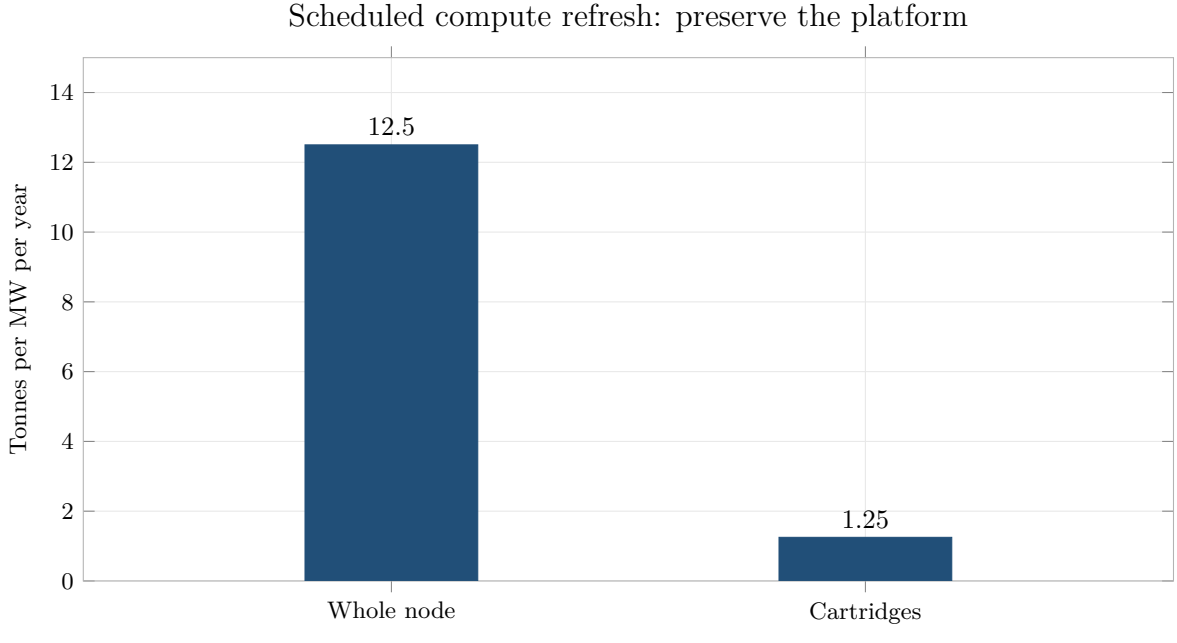
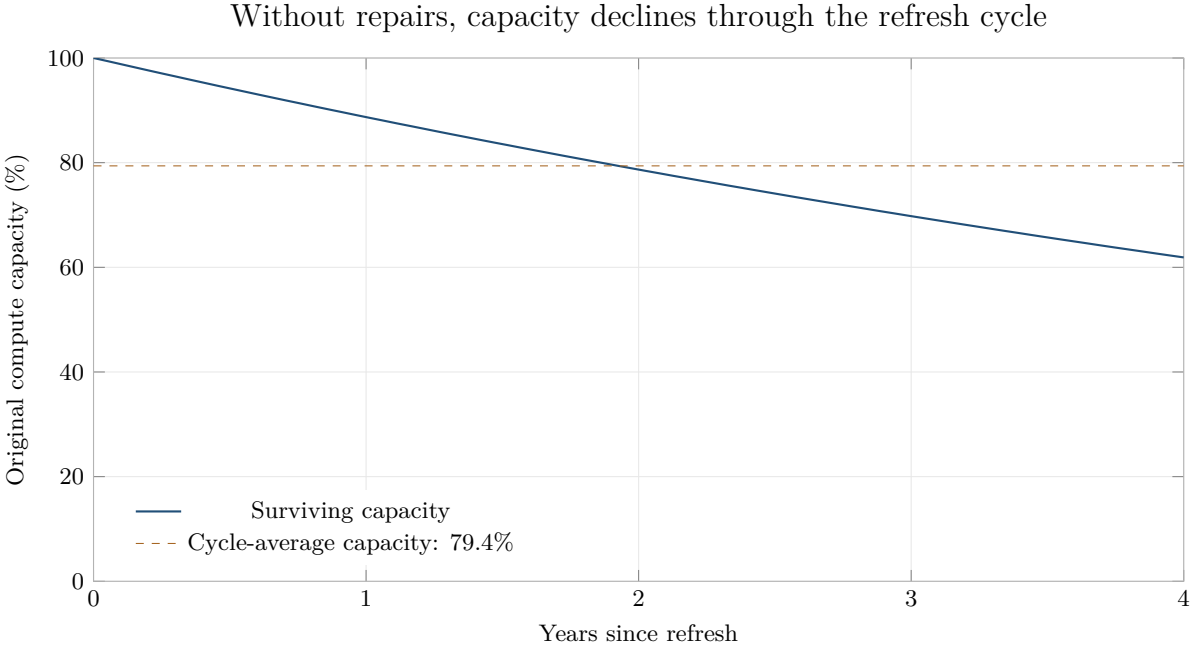
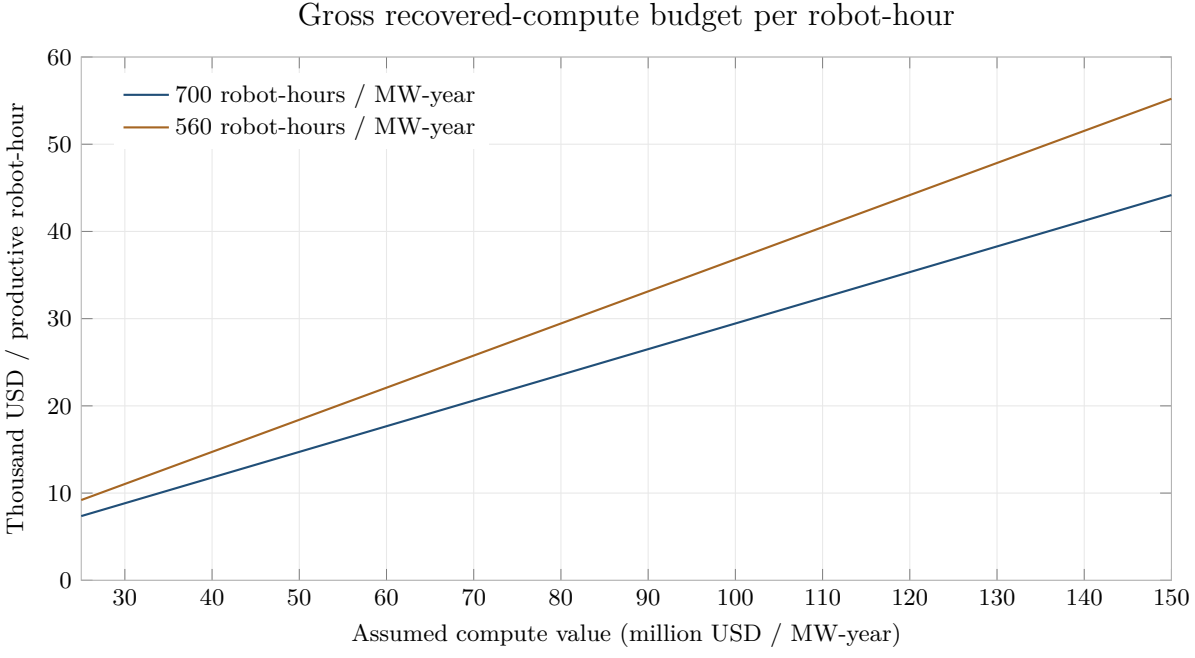

Orbital deployment cost relative to ground
9.7% relative decline
6% relative decline
Ground = 1
Cost ratio (log scale)
10^{0}
10^{-1}
10^{-2}
2026
2036
2046
2056
2066
2076
Year


What growth would reaching 10 MW require?
Doubling every 9 months
Doubling every 12 months
Doubling every 18 months
10 MW reference architecture
Assumed IT capacity, MW (log scale)
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
2026
2031
2036
2041
2046
Year


Scheduled compute refresh: preserve the platform
Tonnes per MW per year
0
2
4
6
8
10
12
14
12.5
1.25
Whole node
Cartridges


Without repairs, capacity declines through the refresh cycle
Original compute capacity (%)
100
80
60
40
20
0
0
1
2
3
4
Years since refresh
Surviving capacity
Cycle-average capacity: 79.4%


Gross recovered-compute budget per robot-hour
700 robot-hours / MW-year
560 robot-hours / MW-year
Thousand USD / productive robot-hour
Assumed compute value (million USD / MW-year)
0
10
20
30
40
50
60
30
40
50
60
70
80
90
100
110
120
130
140
150